

SANA NIROOMAND

Computer Engineering Student | AI/ML, Biomedical AI, Drug Design & Startup Engineering

✉ sana.niroomand@gmail.com ☎ +98 990 171 72 32 📍 Tehran, Iran in Sana Niroomand 🌐 SanaNiroomand

EDUCATION

B.S. in Computer Engineering

Sharif University of Technology

📅 Sept 2023 – Present 📍 Tehran, Iran

- GPA: 18.88/20.0 (3.78/4.0)
- Coursework: ML, AI, Biomedical Image Analysis, Computational Drug Design, Bioinformatics, Algorithms, Data Structures, Linear Algebra, Probability.

WORK

Research Intern – HDR Object Detection

Middle East Technical University (METU)

📅 Jul 2026 – Present 📍 Ankara, Turkey

- Conducting research under the supervision of Prof. Aydin Alatan to determine whether object detection performs better directly on HDR data or after HDR-to-LDR tone mapping.
- Designing and running comparative experiments on the local HDR4RTT dataset to evaluate how tone-mapping stages affect object-detection performance.

AI EXPERIENCE

PI-CAI Prostate Cancer Prediction

Medical Imaging / Weak Supervision / MIL

- Built an interpretable, uncertainty-aware MIL pipeline for prostate cancer prediction using PI-CAI-style MRI data.
- Adapted multi-scale attention-based aggregation for weak supervision where lesion-level annotations are limited.
- Focused on interpretability, uncertainty estimation, and evaluation for medical imaging workflows.

Yara – Intelligent Medical Assistant

Healthcare AI / Clinical NLP / LLM Workflows

📅 Ongoing

- Designing an AI-native medical assistant concept that converts doctor–patient conversations into structured notes, summaries, and searchable patient history.
- Exploring patient-record Q&A workflows over visits, labs, medications, and generated clinical summaries.

Transcriptomic Analysis of Psoriasis and Psoriatic Arthritis Skin

Bioinformatics / RNA-seq

- Reproduced parts of a transcriptomics study using public RNA-seq datasets.
- Worked on differential gene expression analysis and interpretation of immune-related transcriptional patterns.

Chorus – AI-Powered Language Learning Bot

Python / Generative AI / NLP

- Developed a multilingual Telegram bot that transforms songs into adaptive language-learning exercises using the OpenAI and LRCLIB APIs.
- Implemented lyric retrieval, audio metadata extraction, fill-in-the-blank generation, answer evaluation with explanations, beginner word banks, and cultural-context guidance.

RESEARCH & TEACHING

Vision-Language Models for Plant Disease Detection

Deep Learning Lab

📅 Feb 2025 – May 2025

- Explored VLM-based plant disease diagnosis from image prompts under the supervision of Prof. Rohban and Prof. Sabzi.

Teaching Assistant – Advanced Programming

OOP and Design Patterns

📅 Sharif University of Technology

- Assisted students with OOP, design patterns, project structure, debugging, and implementation logic.

ACHIEVEMENTS

- **Top 0.14% Nationally**
Iranian Nationwide University Entrance Exam in Mathematics, among 147,317 participants.
- **Gold Medal**
International exhibition in South Korea for a botanical antimicrobial spray project.
- **2nd Place**
Inventions & Achievements Track at the Avicenna Festival.
- **Provincial Distinguished Rank**
Khwarizmi Youth Festival.

SKILLS

Python

C++

C

Java

PyTorch

NumPy

Pandas

Scikit-learn

Computer Vision

Deep Learning

Biomedical AI

Bioinformatics

Drug Design

Git

Linux

LANGUAGES

- **English:** Fluent
- **French:** Conversational